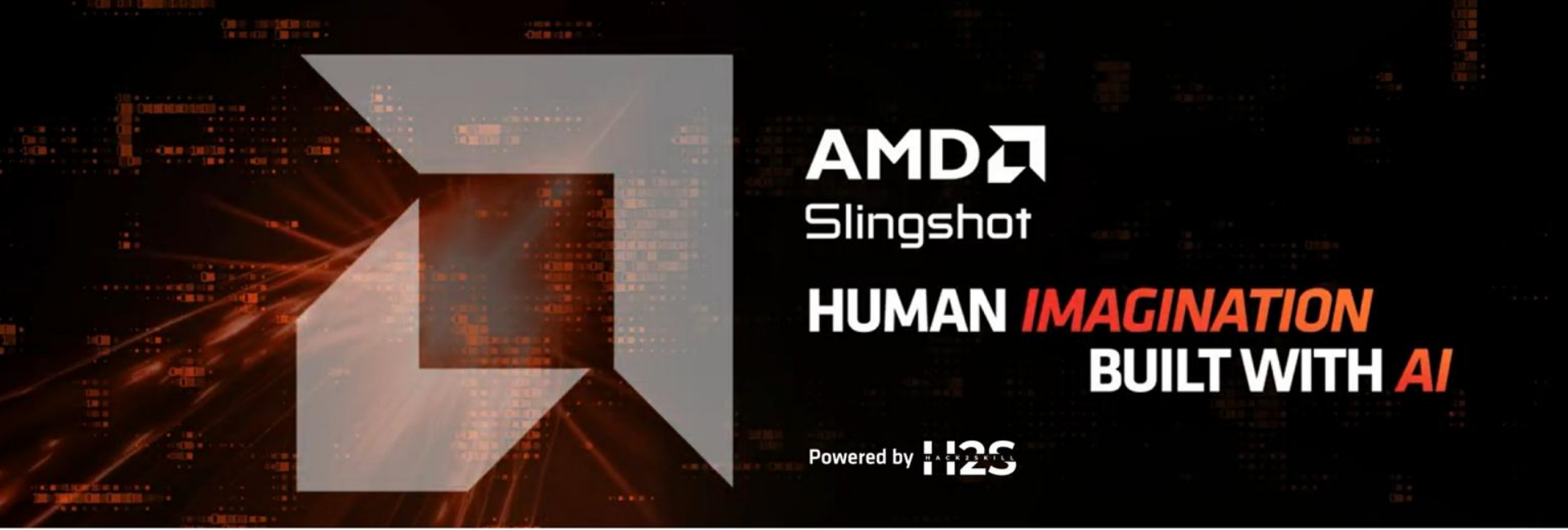


Team Details

- ➔ Team name – Ctrl+Alt+Del
- ➔ Team leader name – Lokesh Kiruala
- ➔ Problem Statement - To develop an AI-powered, on-device skin disease risk screening system that performs real-time image-based analysis using AMD Ryzen AI hardware, enabling early detection support in low-resource and rural settings without relying on constant cloud connectivity.

Brief about the idea

DermaLite AI is a hybrid edge-AI skin disease detection system that analyzes images of skin conditions and predicts possible disease categories along with a risk level assessment. The core image classification model runs fully on-device using AMD-optimized ONNX Runtime for fast and privacy-preserving inference. The system provides explainable results using visual heatmaps and structured medical guidance.

The hybrid approach ensures:

- Local inference for detection (offline capable)
- Optional cloud updates for model improvement
- Privacy-first medical screening support

Opportunities

- **How different is it from any of the other existing ideas?**

  Most dermatology AI tools are cloud-based.
  DermaLite performs on-device inference using AMD Ryzen AI.
  No image upload to external servers (privacy-preserving).
  Includes explainability (Grad-CAM heatmaps).
  Designed for rural accessibility and low-bandwidth environments.

- **How will it be able to solve the problem?**

  Provides early-stage risk screening for common skin diseases.
  Reduces dependency on immediate dermatologist access.
  Encourages early consultation when risk is high.
  Enables health workers to use AI assistance in remote regions.

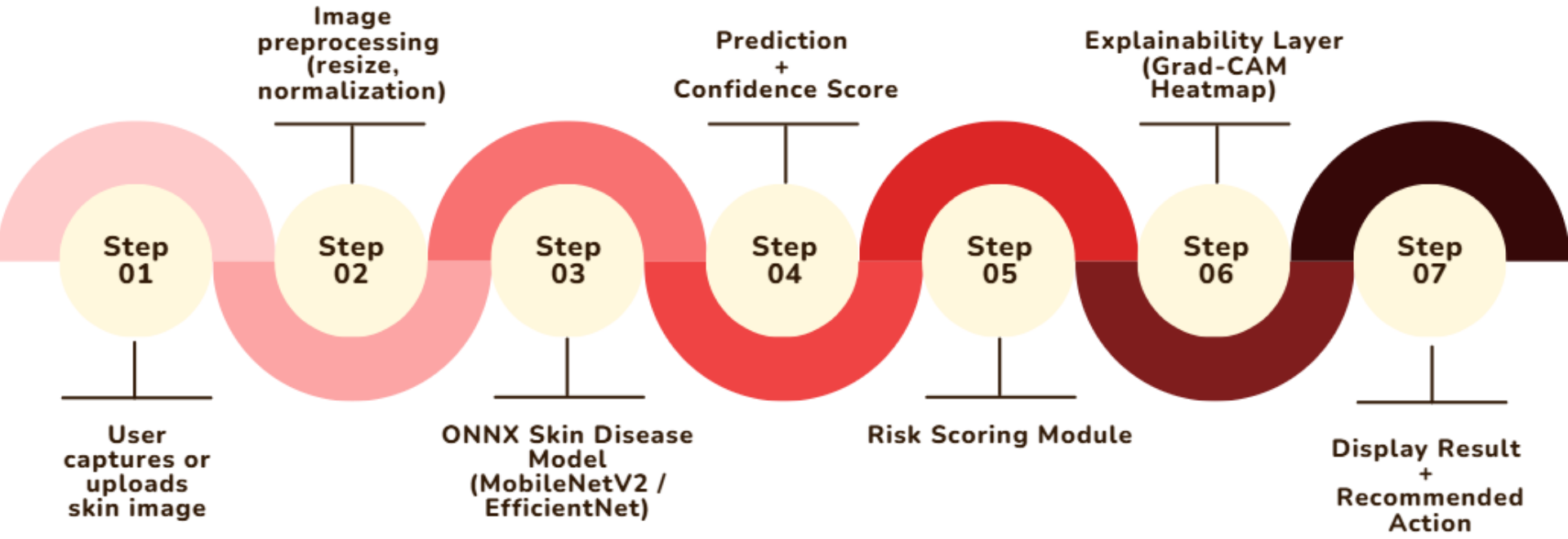
Opportunities

- **USP of the proposed solution**
  - Fully edge-optimized skin detection system
  - ONNX-based AMD hardware acceleration
  - Explainable AI (visual attention maps)
  - Hybrid architecture (offline-first, cloud-optional)
  - Privacy-preserving medical AI

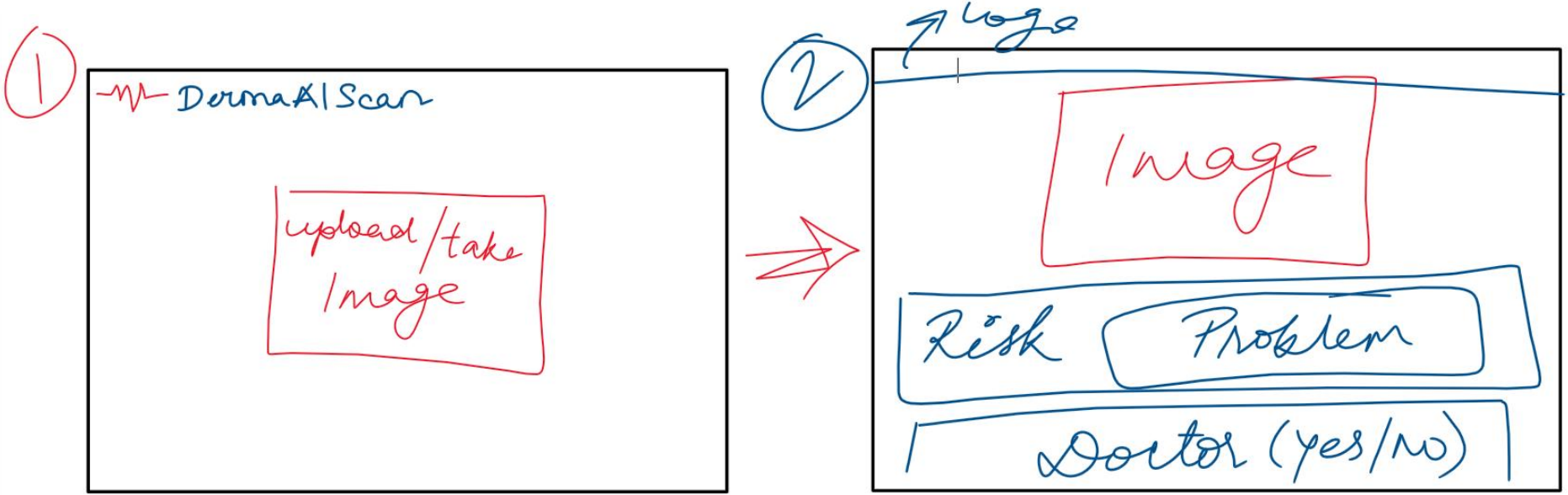
List of features offered by the solution

- Real-time skin image classification
- Risk-level categorization (Low / Moderate / High)
- Confidence score display
- Grad-CAM explainability heatmap
- Offline inference capability
- Local result storage
- Optional medical guidance text explanation
- Privacy-first design (no mandatory cloud upload)

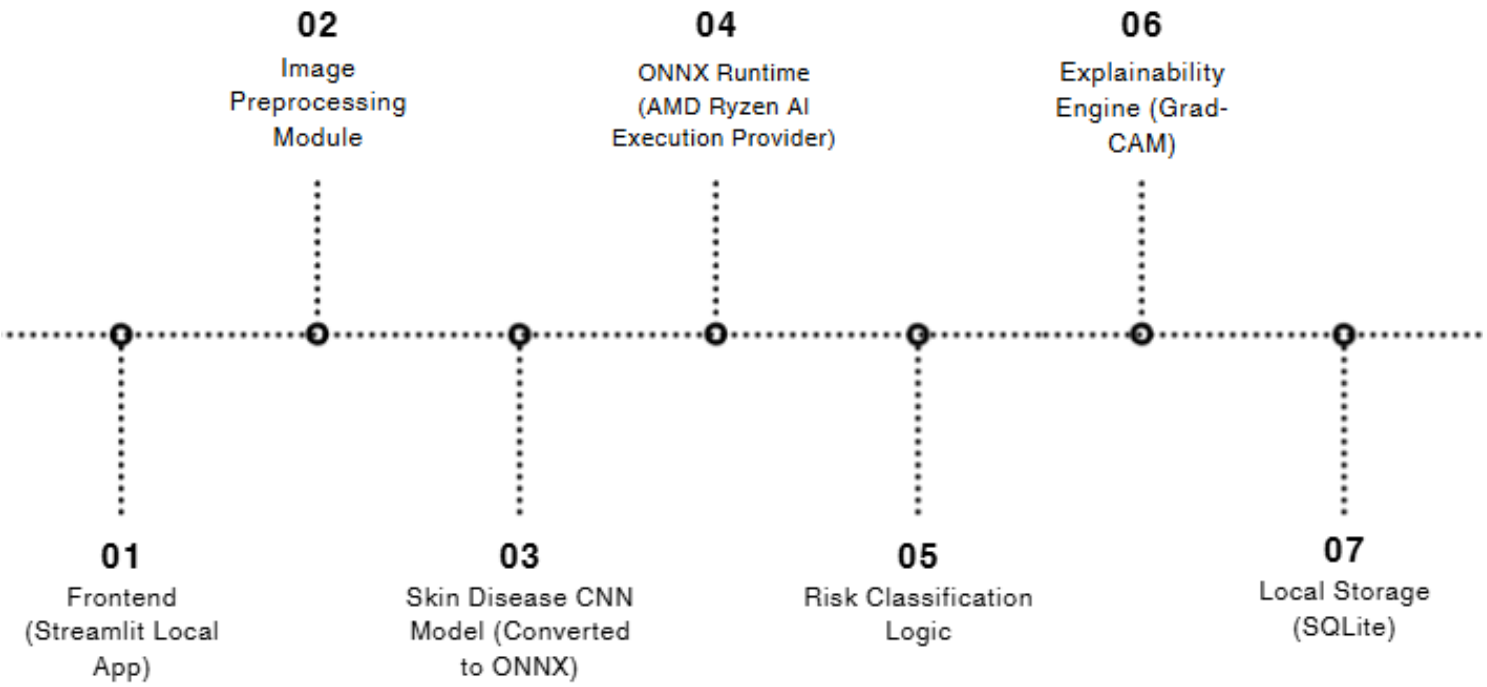
Flowchart


Step 01
User captures or uploads skin image
Step 02
Image preprocessing (resize, normalization)
Step 03
ONNX Skin Disease Model (MobileNetV2 / EfficientNet)
Step 04
Prediction + Confidence Score
Step 05
Risk Scoring Module
Step 06
Explainability Layer (Grad-CAM Heatmap)
Step 07
Display Result + Recommended Action

Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution



Technologies to be used in the solution

COMPONENT	TECHNOLOGY
Image Model	MobileNetV2 / EfficientNet-Lite
Framework	PyTorch
Model Format	ONNX
Inference Engine	ONNX Runtime
UI	Streamlit
Explainability	Grad-CAM
Storage	SQLite
Dataset	HAM10000 / DermNet
OS	Windows 11 / Ubuntu

Usage of AMD Products/Solutions

- Model converted to ONNX for AMD compatibility
- ONNX Runtime with AMD Execution Provider
- Inference optimized for Ryzen AI NPU
- Efficient on-device processing reducing power consumption
- Demonstrates real-world edge AI deployment using AMD hardware

Prototype - Video Link

Google Drive Link-
https://drive.google.com/file/d/1kXYivKxblEk7FhdjxVmdH7z7E9P6vWRJ/view?usp=sharing

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S

Thank you!